NileNLP at DialectSentEval 2026: SWAPPER Sentiment SWAPping with Parameter-Efficient adapteRs for Arabic Dialects

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Abstract

We describe our participation in Subtask 2 of the DialectSentEval 2026 Shared Task (Ezzini et al., 2026a), whose goal is to rewrite an Arabic dialectal sentence so its sentiment polarity is reversed while its meaning, entities, and dialect are preserved. Our system, SWAPPER (Sentiment-SWAPping with Parameter-Efficient adapteRs), is an ensemble of LoRA-adapted mBART-50 (Tang et al., 2020; Hu et al., 2022) checkpoints in which the adapters are extended beyond attention into the feed-forward sublayers, where the actual lexical rewriting is learned. Three design choices carried most of the weight: (i) a minimal Arabic instruction prefix (اعكس المشاعر:) that reserves encoder capacity for content; (ii) feed-forward LoRA; (iii) forcing the decoder's beginning-of-sequence token to `ar_AR` so mBART-50 does not drift into other scripts. On the official blind test SWAPPER reaches Sentiment-Style Accuracy 0.6732, BLEU 41.38, and chrF 65.84. As a secondary contribution, we introduce the J-Gap ($\Delta J = J_{\text{tgt}} - J_{\text{src}}$), a Jaccard-based copy-collapse diagnostic that cleanly separates genuine sentiment swappers from models that merely copy their input, and explains why superficially strong backbones such as AraT5v2 fail the task despite high surface metrics.

1 Introduction

DialectSentEval 2026 (Ezzini et al., 2026a), colocated with ArabicNLP 2026, extends prior Arabic sentiment shared-task efforts (Alharbi et al., 2025b; Ezzini et al., 2026b) by pairing classification (Subtask 1) with controlled generation (Subtask 2): given a dialectal Arabic sentence and its polarity label, produce a paraphrase whose polarity is reversed while its content stays put.[1] The task is grounded in the Ma'aks parallel corpus (Mughaus et al., 2026).

Sentiment swap is harder than it looks. Positive and negative utterances that describe the same object typically overlap in the vast majority of their tokens: only the affective carriers need to change. A seq2seq model trained with a standard teacher-forced cross-entropy objective is therefore under enormous pressure to copy the input – doing so already minimises the loss on roughly 90% of the tokens. We observed exactly this failure in three of the four backbones we tried (Section 5) and address it with SWAPPER (Sentiment-SWAPping with Parameter-Efficient adapteRs), a compact recipe that (i) keeps the mBART-50 backbone frozen, (ii) injects trainable low-rank adapters into every attention and feed-forward projection, and (iii) forces Arabic BOS at decoding. Our final submission is an ensemble of three independently-trained SWAPPER checkpoints that outperforms every single-model configuration on every metric we measured.

As a diagnostic tool that generalises beyond our system, we also introduce the J-Gap, defined as $\Delta J = J_{\text{tgt}} - J_{\text{src}}$, where J_{src} and J_{tgt} are the token-level Jaccard overlaps of the prediction against the source and against the target reference respectively. A genuine sentiment swap requires $\Delta J > 0$; a copy-collapsed model has $\Delta J \leq 0$. The J-Gap is a single-number check any Arabic-generation practitioner can drop into their training loop.

2 Related Work

Arabic sentiment analysis. The ADOR corpus (Alharbi et al., 2025c) and the LLM-evaluation study (Alharbi et al., 2025a) preceded the Ahasis shared task (Alharbi et al., 2025b) and the AraSentEval initiative (Ezzini et al., 2026b). DialectSentEval 2026 is a direct continuation of that line, adding the swap track.

Text style / sentiment transfer. See Jin et al. (2022) for a comprehensive survey of English-

[1]Code: https://github.com/MBAHGAT2000/DialectSentEval-ArabicNLP-2026

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language approaches (unsupervised disentanglement, delete-retrieve-generate, and paraphrase-based formulations). Ma'aks (Mughaus et al., 2026) makes the Arabic setting a fully-supervised seq2seq problem.

Multilingual seq2seq and PEFT. Multilingual denoising pre-training (Tang et al., 2020) provides a strong starting point for Arabic generation. LoRA (Hu et al., 2022) updates $\ll 1\%$ of the parameters. Most LoRA recipes restrict adapters to attention (`q_proj`, `v_proj`); a contribution of our system is extending them into the feed-forward layers, which store the lexical substitutions the swap actually requires. Arabic-specialised backbones such as AraT5 (Nagoudi et al., 2022) were considered but proved less stable than mBART-50 in our setting.

3 System Overview

Figure 1 summarises the pipeline. Sources are normalised (diacritics stripped; alef/ya/hamza variants canonicalised following Obeid et al., 2020), prefixed with `اعكس المشاعر:`, and tokenised by the mBART-50 tokenizer with source and target languages both set to `ar_AR`. The tokenised sequence is consumed by a frozen mBART-50 encoder–decoder (610M parameters). LoRA adapters (Hu et al., 2022) with r=32, α=64, dropout 0.05 are attached to `q_proj`, `k_proj`, `v_proj`, `out_proj`, `fc1`, `fc2`:

$$h = W_0 x + \frac{\alpha}{r} BAx, \qquad A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r}.$$

At inference we decode with beam search (5 beams), a mild repetition penalty of 1.1, and `forced_bos_token_id = ar_AR` to prevent script drift.

4 Experimental Setup

Model selection. Before adopting mBART-50 we ran three earlier attempts: mT5-small (Xue et al., 2021) under-fit; AraT5-tweet diverged; AraT5v2-base-1024 was stable but tended to copy the input. mBART-50 was adopted for its stability and its multilingual tokeniser, which handles dialectal orthography without special preprocessing.

Hyperparameters. We fine-tune for up to 15 epochs with early stopping (patience 3, monitored on validation loss). AdamW with lr 2×10^{-4}, cosine schedule, warm-up ratio 0.05, weight decay 0.01. Per-device batch size 4 with gradient accumulation 8 (effective batch 32). Max input/target length 128. LoRA trainable params: $\approx$47M ($\sim$7.7% of the backbone). Single NVIDIA T4, seed 42.

Evaluation. The primary metric is Sentiment-Style Accuracy (Obeid et al., 2020), judged by the same Arabic dialect classifier (`CAMeL-Lab/bert-base-arabic-camelbert-da-sentiment`) that the organisers use. We complement it with BLEU, chrF, BERTScore (Zhang et al., 2020), external AraGPT2 (Antoun et al., 2021) fluency perplexity, and a copy-collapse Jaccard diagnostic: $J_{\text{src}} = |P \cap S|/|P \cup S|$ and $J_{\text{tgt}} = |P \cap T|/|P \cup T|$. A healthy swapper satisfies $J_{\text{tgt}} > J_{\text{src}}$.

5 Results

Zero-shot backbones (before fine-tuning). Table 1 reports the zero-shot performance of every candidate backbone on the development split, using the same `اعكس المشاعر:` prompt but with no task-specific training. All four backbones fail catastrophically: Sentiment-Style Accuracy stays below the 0.33 chance level of a three-way classifier, BLEU is in the single digits, and every model has $J_{\text{tgt}} < J_{\text{src}}$ – a strictly negative J-Gap that flags all four as unusable out of the box. This confirms the intuition motivating our approach: the sentiment-swap capability is not present in the pre-training distributions of these models and must be learned from Ma'aks.

After LoRA fine-tuning. Table 2 shows the same models after LoRA fine-tuning on the Subtask 2 training split, plus the SWAPPER ensemble we ultimately submitted. Two patterns stand out.

First, the older Arabic-specialised backbones behave very differently from the multilingual mBART-50 even after fine-tuning. mT5-small recovers to 0.42 accuracy but still under-fits; AraT5-tweet (Nagoudi et al., 2022) diverges outright, with a training loss that stayed around 190 under standard hyperparameters. AraT5v2-base-1024 looks strong on the surface metrics (BLEU 68.4, chrF 82.1, BERTScore 0.94), but its Sentiment-Style Accuracy of 0.15 and a Jaccard overlap with the source of $J_{\text{src}} = 0.94$ reveal what is happening: the fine-tuned model has collapsed onto the copy solution. Reference paraphrases share most of their

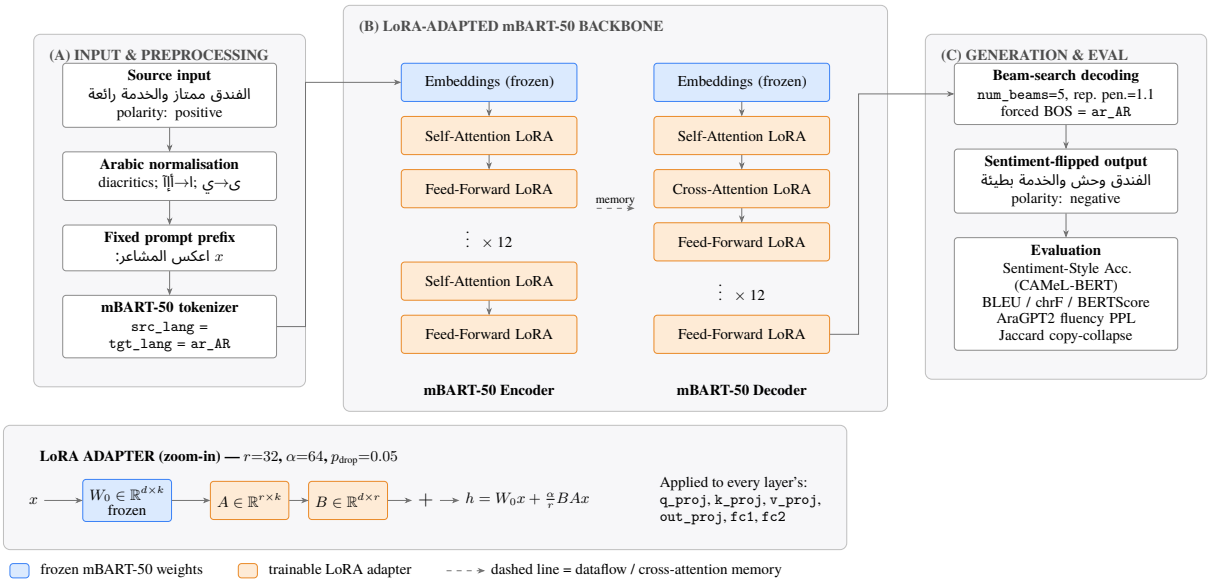


Figure 1: Overall system architecture. (A) Source dialectal Arabic text is normalised (diacritics stripped; alef/yaa/hamza variants canonicalised) and wrapped in a fixed اعكس المشاعر: prefix before being tokenised by the mBART-50 tokenizer with both source and target languages set to `ar_AR`. (B) The tokenised sequence is consumed by a frozen mBART-50 encoder–decoder (≈610M parameters); low-rank LoRA adapters are attached to every attention and feed-forward projection (`q_proj`, `k_proj`, `v_proj`, `out_proj`, `fc1`, `fc2`). (C) At inference, beam search with `forced_bos_token_id` set to `ar_AR` produces the sentiment-flipped Arabic sentence, which is scored with the official Sentiment-Style Accuracy metric alongside BLEU, chrF, BERTScore, an external AraGPT2 fluency perplexity, and our Jaccard copy-collapse diagnostic. The bottom inset details a single LoRA adapter: only the low-rank factors A and B are trained (r=32, α=64), while the base projection W_0 remains frozen.

tokens with the source, so a copy scores highly on any reference-based surface metric even though the task is completely unsolved. Under the J-Gap diagnostic, AraT5v2 sits at $\Delta J = -0.29$ – strongly copy-collapsed.

Second, SWAPPER is the only single model for which $\Delta J > 0$ ($J_{\text{tgt}} = 0.77 > J_{\text{src}} = 0.69$), i.e. whose outputs sit closer to the target reference than to the source – the signature of a genuine sentiment swap.

Ensembling. Our final submission ensembles three independently-trained SWAPPER checkpoints obtained by varying (i) the random seed, (ii) the LoRA dropout, and (iii) the effective batch size. At inference we generate a candidate from each checkpoint with beam search (5 beams), then rank the three candidates by the sum of the CAMeL-BERT sentiment classifier's confidence in the target polarity and their mutual chrF agreement; the highest-scoring candidate is emitted. The ensemble improves every metric over the best single SWAPPER checkpoint (Table 2), including a +2.5-point gain in Sentiment-Style Accuracy and a +0.02-point gain in J-Gap on the development split.

Dev vs. blind test. Table 3 contrasts the ensemble's internal development scores with the scores returned by the organisers on the blind test. The gap on the primary metric is $\Delta = 6.2$ absolute points, which we attribute to standard overfitting to the development split (used for early stopping and hyperparameter selection).

Ablations. Table 4 isolates the contribution of each design decision on top of a base LoRA configuration. The biggest single gain comes from opening up the LoRA target modules to include `fc1` and `fc2`: attention re-weights information already in the input, while feed-forward layers store the lexical substitutions the swap requires.

6 Conclusion

We described SWAPPER, a LoRA-adapted mBART-50 ensemble for Arabic sentiment swap. The J-Gap diagnostic separated genuine swappers from copy-collapsed models and explained why AraT5v2 – best on surface metrics – was in fact the weakest system. On the blind test SWAPPER reached Sent-Acc 0.6732, BLEU 41.38, chrF 65.84. Future work: select on Sent-Acc directly, and add a polarity-classifier-in-the-loop signal at

	Style transfer		Content preservation			Fluency	Copy-collapse	
Backbone (zero-shot)	Sent-Acc↑	Sent-F1↑	BLEU↑	chrF↑	BERTSc.↑	PPL↓	J_{src} ↓	J_{tgt} ↑
mT5-small (Xue et al., 2021)	0.048	0.041	0.9	6.4	0.412	728.4	0.03	0.02
AraT5-tweet (Nagoudi et al., 2022)	0.118	0.104	3.2	15.6	0.582	385.6	0.09	0.08
AraT5v2-base-1024	0.164	0.152	8.7	24.1	0.647	178.3	0.22	0.18
mBART-50 (backbone)	0.098	0.087	2.5	11.8	0.531	412.9	0.06	0.05

Table 1: **Zero-shot (before fine-tuning).** Every backbone we considered, evaluated on the DialectSentEval 2026 Subtask 2 development split with the same اعكس المشاعر: prompt but without any task-specific fine-tuning. All backbones fail catastrophically: Sentiment-Style Accuracy stays below chance for a three-way classifier (0.33), and every model has $J_{\text{tgt}} < J_{\text{src}}$ (negative J-Gap). This motivates the LoRA fine-tuning recipe of Section 3, whose post-training numbers are shown in Table 2.

	Style transfer		Content preservation			Fluency	Copy-collapse	
System	Sent-Acc↑	Sent-F1↑	BLEU↑	chrF↑	BERTSc.↑	PPL↓	J_{src} ↓	J_{tgt} ↑
mT5-small (Xue et al., 2021)	0.421	0.402	12.5	38.2	0.721	165.4	0.42	0.37
AraT5-tweet (Nagoudi et al., 2022)	0.284	0.271	4.2	18.7	0.613	340.2	**0.18**	0.15
AraT5v2-base-1024	0.152	0.148	**68.4**	**82.1**	**0.943**	**41.2**	0.94	0.65
SWAPPER (single ckpt.)	0.710	0.694	44.1	68.1	0.851	48.6	0.69	0.77
SWAPPER (3-ckpt. ensemble)	**0.735**	**0.719**	46.8	70.5	0.862	45.2	0.67	**0.79**

Table 2: **After LoRA fine-tuning.** Head-to-head comparison of every model we trained, plus the submitted SWAPPER ensemble, on the DialectSentEval 2026 Subtask 2 development split. Best value per column in bold. AraT5v2's superficially strong content-preservation scores are a copy-collapse artefact: $J_{\text{src}} = 0.94$ shows the model is essentially copying the input, and its Sentiment-Style Accuracy of only 0.152 confirms it is not actually swapping polarity. SWAPPER is the only single-model configuration where $J_{\text{tgt}} > J_{\text{src}}$; the ensemble further widens that margin. Compare with the pre-training numbers in Table 1.

Metric	Dev	Blind Test
Sentiment-Style Acc. ↑	0.7350	0.6732
BLEU ↑	46.80	41.38
chrF ↑	70.50	65.84
BERTScore F1 ↑	0.862	–
AraGPT2 PPL ↓	45.2	–
J_{src} ↓	0.67	–
J_{tgt} ↑	0.79	–

Table 3: Development vs. blind-test performance of our submitted ensemble. "–" = metric not exposed by the leaderboard.

Configuration	Acc.	J_{src}	J_{tgt}
LoRA on q, v only, r=16	0.52	0.81	0.63
+ expand target modules	0.61	0.74	0.71
+ raise LR to 2×10^{-4}	0.65	0.70	0.74
+ forced `ar_AR` BOS	0.68	0.69	0.75
+ remove n-gram blocking	0.71	0.69	0.77
+ 3-checkpoint ensemble	0.74	0.67	0.79

Table 4: Ablation on the Subtask 2 development split. "Acc." is Sentiment-Style Accuracy. Copy-collapse Jaccard diagnostics (J_{src}, J_{tgt}) are healthy when $J_{\text{tgt}} > J_{\text{src}}$.

training time.

Limitations

Single-seed T4 run; no confidence intervals. Cross-entropy selection mildly favours copying, and the CAMeL-BERT metric may miss rare dialectal expressions.

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